Title

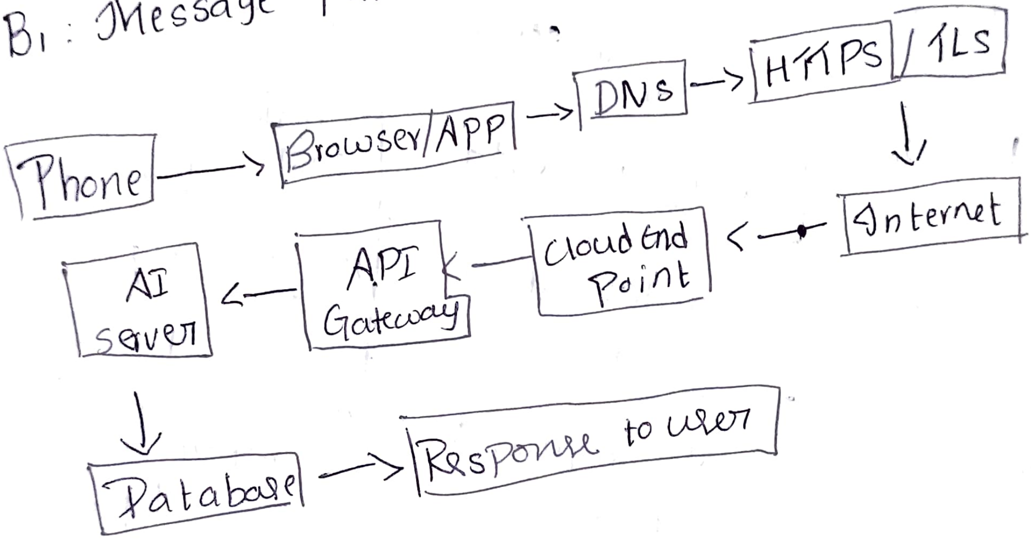
A1 Cloud-Based Crop water AI: Irrigation scheduling and water Stress

Main user Action:- Farmer requests an irrigation Schedule and water Stress Prediction for crops

A2: Request Sentence: "Analyze my crop data, Predict water stress, and provide the best irrigation schedule"

Part B - Network Journey

B1: Message Path



Part C - Cloud Landing Zone

Cloud Need	Component choice	Services Mobile	Justify
user login	Authentication service	Saas	secure user Login
APP Backend	AI Prediction server	Paas	Runs Irrigation AI model

Database	Cloud database	DBaas	stores far mer & crop da
File storage	Cloud storage	Iaas	stores images & reports
Monitoring	Cloud monitoring	SaaS	Tracks Performance and errors

B2 Technical Translation Table

Layer/ Component	what it does	Possible Failure	Cloud-side Protection
DNS	Finds the cloud server adress	DNS failure	Backup DNS
HTTPs/TLs	Encrypts data	certificate error	SSL/TLS certificates
TCP/IP	Transfers data reliably	Packet loss	Retransmission
outer/ISP/ Internet	sends request to cloud	Network congestion	Multiple network Paths
API ateway	Receives and routes requests	Server overload	Load balancer & auto scaling

Part D - API contract

- End Point: /api/irrigation/predict
- HTTP method: POST
- Response Data: water stress level, Irrigation Time, water Quantity
- Authentication Rule: login using JWT Token
- Error case: Invalid data or server unavailable

Part E - Elasticity and Scalability

E1 - Peak Requests/Minute: 7200 ÷ 10 = 720 requests/minute.

E2 - Required Capacity (25% spare): 720 × 1.25 = 900 requests/minute

E3 - Number of Instances: 900 ÷ 180 = 5 insta

E4 - Elastic Design Decision: use Auto Scaling to automatically add or remove cloud servers based on traffic

E5 - Measured service Metrics

- Request count
- Response time

Part F – AI / Google Research Evidence

- Search / Prompt: Cloud AI irrigation scheduling using machine learning

- Tool used: chatGPT
- useful Finding: AI Predicts crop water needs usin weather & soil data

- How I applied it: I used AI to recommend irrigati schedules and detect water stress for farmers